A Quantum Energy Trading solution

Ismael Ahrazem Dfuf

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Abstract

As new Distributed Energy Resources (DERs) continue to proliferate in the electricity market, their integration into the distribution network presents both opportunities and challenges. When optimally managed, DERs such as residential batteries, electric vehicles, and commercial energy storage systems can improve grid stability, enhance operational efficiency, and provide economic value to the stakeholders that deploy them. However, their inherently decentralized and dynamic nature introduces significant complexity into grid planning and market participation strategies.

The electricity market is unique in terms of commodity trade, as electricity cannot be economically stored at large scales over long durations. This necessitates continuous balancing between generation and consumption. This work proposes an energy trading solution designed to manage the charging, idling, and discharging cycles of an ecosystem of Commercial/Industrial (C/I), Residential and EV batteries using a combinatorial optimization approach based on Adiabatic Quantum Computing (AQC). The core objective is to simulate and optimize the operational decisions of these DERs. The proposed solution addresses a Multi-Objective Optimization problem using AQC and the QUBO framework, where we aim to optimize (a) Flattening demand through Peak-Valley control minimization to avoid stress on electric infrastructure, (b) Electricity cost minimization, and (c) Battery readiness for emergency dispatch.

This work demonstrates how emerging quantum optimization techniques can potentially support the efficient scheduling of DERs in the presence of uncertain market conditions and operational constraints. The application of quantum combinatorial optimization represents a hybrid framework tailored for large scale and complex modern decentralized energy systems.

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Acronyms

DER Distributed Energy Resource

QUBO Quadratic Unconstrained Binary Optimization

AQC Adiabatic Quantum Computing

Chapter 1

Objectives

The main objective of this project is to develop a Quantum-based optimizer that coordinates the energy trading operations of an ensemble of batteries. Specifically, the work focuses on simulating the optimal Charging, Idling, and Discharging behavior of EV, Residential and Commercial/Industrial batteries using a combinatorial optimization framework implemented through Adiabatic Quantum Computing (AQC). While the full solution is envisioned to integrate predictive components—including (1) a Quantum Machine Learning (QML) binary classifier to detect extreme electricity demand events, and (2) a QML-based time series forecasting module to predict electricity prices and demand—these predictive modules are left for future development. In this release, the emphasis is placed on constructing a multi-objective Quantum optimizer that simultaneously minimizes peak-valley load disparities in the grid, reduces trading costs, and ensures sufficient battery state-of-charge for emergency readiness.

Chapter 2

Introduction

2.1 Introduction

The electricity market is fundamentally distinct from other commodity markets due to the technical difficulty of storing electricity at scale. This limitation necessitates a continuous and precise balancing of supply and demand across the grid. The growing penetration of Distributed Energy Resources (DERs)—such as electric vehicle (EV) batteries, stand-alone commercial batteries, and residential solar-plus-storage systems—introduces new challenges and opportunities in achieving this balance efficiently and economically.

Battery Energy Storage (BES) systems primarily generate revenue through energy arbitrage, taking advantage of price fluctuations across different market intervals. However, as BES deployment scales and ancillary service markets become increasingly saturated, operators must explore more innovative strategies to capture additional value. One such strategy is DART optimization (Day-Ahead and Real-Time), which enables participation in both scheduling and real-time adjustment markets to exploit temporal arbitrage opportunities.

In this work, we propose an Energy Trader solution that provides a simulation and optimization framework for managing an ecosystem of DERs with surplus stored energy. The proposed solution enables coordinated scheduling of charging, discharging, and idling operations across an aggregation of heterogeneous batteries—ranging from residential and EV storage to commercial/industrial (C/I) systems. The goal is to develop an operational strategy that maximizes economic return while satisfying grid reliability and emergency preparedness constraints.

1. **Energy arbitrage**: management of electricity prices volatility to profit financially by selling when prices are high and buying at low prices.

2. **Peak-Valley Control Optimization**: Load variance reduction optimization. The goal is to flatten the net load variability [1] in order to reduce the stress on the electric infrastructure. This is achieved by:
   - **Valley Filling:** Batteries charge during low-demand/high-renewable periods. In this case we are absorbing excess generation that would otherwise be curtailed (wasted).
   - **Peak Shaving:** Batteries discharge during high-demand/low-renewable periods (e.g., evening grid peaks).

3. **Emergency Reserve Requirement:** The final stored energy in each battery must be at least $\beta\%$ of its total capacity to ensure emergency readiness and mitigate range anxiety:

$$E_{\text{final}}^{(i)} \geq \beta \cdot C_{\text{battery}}^{(i)} \quad \forall i \in \text{Batteries}$$

The Quantum Energy Trader (QET) system integrates multiple computational modules to simulate and optimize the energy trading operations of a distributed ecosystem of batteries, including residential, electric vehicle (EV), and commercial/industrial (C/I) energy storage systems. The key components of the framework are as follows:

- **Data Source and Ingestion:** Raw grid data (e.g., prices, demand, congestion metrics) is ingested from external sources such as GridStatus.io. This step feeds downstream processes with high-frequency, real-time data streams necessary for forecasting and optimization.

[1] Note: Net Load = Base Load + (DERs Charge - DERs Discharge).

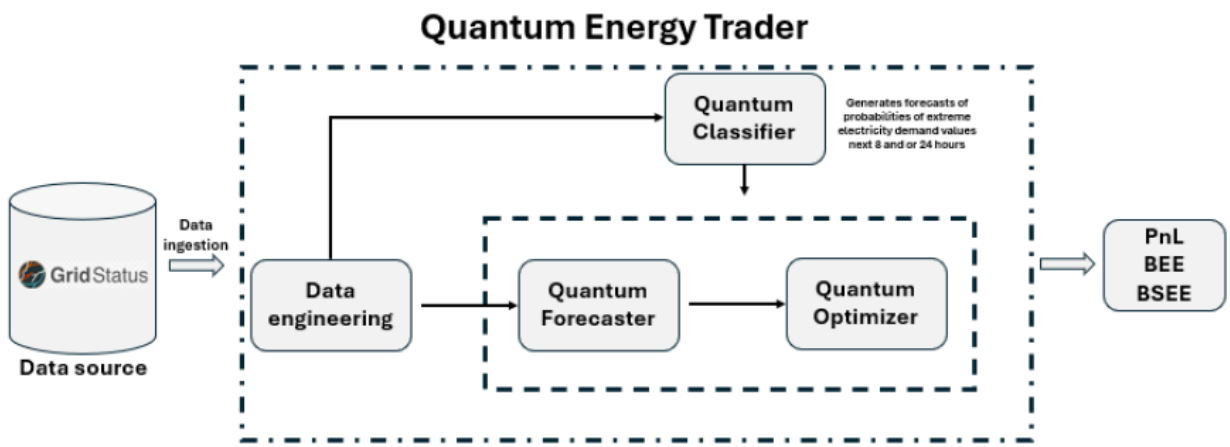


Figure 2.1: Architecture of the Quantum Energy Trader (QET) framework.

- **Data Engineering:** This module preprocesses and transforms the ingested data into structured features suitable for machine learning models. It includes temporal feature extraction, normalization, windowing, and handling of missing data.
- **Quantum Classifier (Simulated/Future release):** A binary classification model estimates the probability of extreme electricity demand or price events within upcoming trading windows (8 and 24 hours). These events typically correspond to periods of peak grid stress or price volatility—ideal conditions for battery discharge. By anticipating such events, the system ensures that stored energy is deployed when its economic value is highest. This not only maximizes revenue but also reduces battery cycling during low-value periods, helping preserve battery health and longevity. The classifier is implemented using classical machine learning in this version, but is intended to be replaced by a quantum classifier in future releases.
- **Quantum Forecaster (Simulated/Future release):** This module predicts electricity prices and demand profiles using time series forecasting. These predictions provide the necessary forward-looking context for optimization. Like the classifier, the forecaster is currently based on classical ML, with a future upgrade path toward quantum time series prediction models.
- **Quantum Optimizer:** The core component of the QET framework and module developed in this work. It formulates a combinatorial optimization problem to determine optimal charging, discharging, and idling actions across the battery fleet. The problem is expressed as a Quadratic Unconstrained Binary Optimization (QUBO) model and solved using Adiabatic Quantum Computing (AQC). This allows the system to efficiently explore large solution spaces constrained by grid conditions, pricing signals, and battery health requirements.
- **Output Metrics:** The framework returns profit and loss (PnL), balance energy exchange (BEE), and battery system energy evolution (BSEE) as key performance indicators.

Why Use Adiabatic Quantum Computing (AQC)?

The optimization problem at the heart of this system is inherently combinatorial: it involves binary decision variables for each battery's operation at each timestep, subject to multi-dimensional constraints (e.g., capacity, energy price windows, emergency readiness). Classical solvers for such problems often suffer from exponential growth in compute time and memory. AQC—by leveraging quantum tunneling and superposition—is capable of exploring large solution spaces more efficiently, particularly for problems well-suited to QUBO encoding.

Why Combine Classical ML with Quantum Optimization?

Forecasting tasks like time series prediction and classification are currently more practical to perform with classical machine learning techniques due to tool maturity, data volume, and quantum hardware limitations. However, integrating these predictive models with a quantum optimizer enables a powerful hybrid workflow: data-driven

insights from ML inform the optimization layer, while quantum processing accelerates the most computationally intensive part—decision making. This architecture serves as a stepping stone toward fully quantum pipelines as Quantum Machine Learning (QML) tools mature. A future release of this work will be the development of a fully quantum energy trader where all components will leverage quantum capabilities. The combination of classical machine learning for forecasting with quantum combinatorial optimization represents a hybrid framework tailored for the complexities of modern decentralized energy systems.

The proposed Quantum Energy Trader (QET) framework has the potential to bring significant value across a broad spectrum of energy market stakeholders. Utility companies can benefit from improved load balancing and grid stability by coordinating distributed energy resources (DERs) such as residential batteries, electric vehicles (EVs), and commercial/industrial (C/I) storage assets. Retail energy providers and energy traders gain access to a new class of high-frequency, data-driven arbitrage strategies that can increase profit margins in real-time and day-ahead markets. For prosumers and facility managers, especially those operating large C/I batteries, this solution can reduce electricity bills through intelligent scheduling aligned with price signals and peak demand forecasts. Regulatory bodies and transmission system operators (TSOs) may also leverage such optimization frameworks to support demand response programs and grid reliability during periods of extreme load.

A key contribution in this work lies in the use of Adiabatic Quantum Computing (AQC) to solve the underlying combinatorial optimization problem formulated as a Quadratic Unconstrained Binary Optimization (QUBO) model used for energy trading. Traditional approaches based on Mixed Integer Linear Programming (MILP) or heuristic algorithms often struggle to scale efficiently as the number of DERs, time intervals, and operational constraints increases. In contrast, AQC leverages quantum tunneling and superposition to explore a vast solution space in parallel, making it a promising paradigm for large-scale binary decision problems where classical solvers may face exponential runtimes. By integrating classical machine learning models for forecasting with a quantum optimizer for decision-making, the QET framework achieves a hybrid architecture that is both computationally powerful and practically feasible on today's hardware. This positions the solution as a forward-compatible tool for smart grid planning and high-frequency energy trading applications.

Chapter 3

Related works

3.1 Related works

The work by [3] employed **Deep Reinforcement Learning** (specifically DDQN, DDPG, and P-DQN algorithms) to optimize electric vehicle (EV) charging by maximizing photovoltaic self-consumption and battery state-of-charge. The model-free approach outperformed rule-based and MPC controllers, achieving **95% charging efficiency** while reducing grid dependence, demonstrating RL's potential for real-time energy management in smart grids. The study by [7] developed a **Q-learning** algorithm for energy storage arbitrage in real-time electricity markets, introducing a novel reward function incorporating historical price averages. This approach achieved **4.8× higher profits** compared to baseline methods by effectively learning optimal charge/discharge timing without price forecasting, demonstrating RL's superiority for stochastic market conditions. The research by [1, 5] proposed **convex optimization-based control strategies** for battery energy storage in data centers, achieving **31% reduction in electricity costs** through peak shaving. Two approaches were developed: an optimal full-horizon method with complete demand knowledge and a practical limited-horizon algorithm using short-term predictions, both accounting for battery conversion/leakage losses. Evaluations on Google data center traces showed lithium-ion batteries provided the best performance among realistic storage options. This study [6] aims to maximize profits for Battery Energy Storage Systems (BESS) operators while maintaining grid stability in distribution networks with high photovoltaic (PV) generation and fast electric vehicle (EV) charging demand. The authors employ the Twin Delayed Deep Deterministic Policy Gradient (TD3) reinforcement learning algorithm to optimize BESS power scheduling, comparing it against Deep Deterministic Policy Gradient (DDPG), Particle Swarm Optimization (PSO), and Simulated Annealing (SA). Results demonstrate TD3's superior performance, achieving a 9.46% Internal Rate of Return (vs. 8.69% for DDPG) while effectively managing voltage constraints and requiring minimal reserve energy (0.492MWh in worst-case scenarios). This study [2] proposes a real-time multi-objective optimization framework for electric vehicle (EV) charging management in residential microgrids with photovoltaic (PV) systems, aiming to balance three conflicting objectives: minimizing electricity costs, reducing battery degradation, and alleviating grid stress while meeting user charging requirements. The authors develop an **Online Multi-Objective Dynamic Programming (OMODP)** algorithm, enhanced with Model Predictive Control (MPC) and Analytical Hierarchy Process (AHP), to efficiently solve the non-convex optimization problem in real-time. Experimental results demonstrate that OMODP outperforms benchmark methods like the Augmented Non-dominated ϵ-Constraint (ANEC) algorithm, achieving **double the end-user profit** while maintaining computational efficiency and successfully validating the approach through hardware implementation with a commercial 18650 battery system.

This paper [4] develops a Grover Adaptive Search (GAS) approach for Constrained Polynomial Binary Optimization problems, demonstrating a theoretical quadratic speedup over classical brute-force search through efficient quantum oracle construction using Quantum Dictionaries ($O(mn^2)$ gates vs. $O(mn^2 \log m)$ for QFT adders). While the method natively handles inequality constraints without classical penalty terms and was validated on IBM's `ibmq_toronto` for small instances, the analysis focuses on quantum feasibility rather than direct performance comparisons with classical optimizers (e.g., branch-and-bound or metaheuristics). The results suggest potential advantages for constraint handling in hybrid quantum-classical algorithms, though practical quantum advantage would require error-corrected hardware and larger-scale implementations to benchmark against classical state-of-the-art solvers.

Chapter 4

Problem formulation

4.1 Problem formulation

In this work, we formulate the energy trading problem as a multi-objective combinatorial optimization task solved under the Quantum Unconstrained Binary Optimization (QUBO) framework. The objective is to determine the optimal charging, discharging, and idle schedules for an ecosystem of distributed batteries—residential, electric vehicle (EV), and commercial/industrial (C/I)—over a predefined time horizon. The formulation balances three key objectives: (1) cost minimization, which aims to reduce energy expenditures by exploiting price arbitrage opportunities; (2) Peak-Valley Control minimization, which encourages operation during off-peak hours to alleviate grid stress; and (3) battery readiness, which ensures a minimum level of stored energy is preserved for emergency use or EV range anxiety.

This optimization is subject to a set of operational and physical constraints, classified as either hard (strict) or soft (penalized). The first constraint is **exclusivity** (hard): a battery cannot be scheduled to charge and discharge simultaneously at any time step. The second is a **consecutive operation constraint** (soft), which discourages back-to-back transitions between charging and discharging states, promoting smoother operational cycles. The third is a **transformer capacity limit** (hard): the total aggregated load—including exogenous demand and battery dispatch—must not exceed the transformer's rated capacity. The fourth is a **minimum final state-of-charge (SOC)** (soft), requiring each battery to retain at least 30% of its energy capacity at the end of the scheduling horizon to account for emergency availability or driving range in the case of EVs.

In the current version of the framework, we assume that all batteries share a fixed final SOC requirement, although in a real-world deployment, each battery owner may specify an individualized SOC target. Similarly, the model assumes a uniform initial SOC across all batteries. These simplifying assumptions, while enabling tractability, may be relaxed in future iterations. A key risk of the proposed solution lies in its reliance on accurate predictions from the upstream machine learning modules, which provide electricity demand forecasts and extreme event classifications. Moreover, because QUBO formulations require the transformation of constraints into unconstrained objectives via penalization terms, careful tuning of these penalties is necessary. Another non-trivial challenge is the multi-scale nature of the problem: objectives and constraints often operate on different numerical scales, requiring appropriate normalization to ensure a fair and balanced contribution to the global cost function. This normalization is crucial for selecting optimal weights and penalty multipliers that preserve the physical and economic intent of the original formulation.

Several quantum and quantum-inspired hardware platforms have emerged in recent years to address combinatorial optimization problems, each with different architectural paradigms and programming interfaces. Among the most widely adopted is **D-Wave Systems**, which offers a commercially available adiabatic quantum computer capable of solving Quadratic Unconstrained Binary Optimization (QUBO) problems natively via quantum annealing. D-Wave also provides an open-source simulator called `dimod`, which enables users to prototype QUBO models before submitting them to actual quantum hardware. Another prominent approach is IBM's **Qiskit Optimization** module, which supports gate-based quantum algorithms such as the Quantum Approximate Optimization Algorithm (QAOA). While Qiskit-based solutions are promising for hybrid quantum-classical workflows, their scalability remains limited by current hardware qubit counts and noise levels.

In this work, we employ the **Fujitsu Digital Annealer (DADK)**, a quantum-inspired digital architecture

specifically designed for large-scale QUBO optimization. Unlike traditional quantum hardware, DADK uses a specialized CMOS-based architecture to emulate quantum annealing behavior at room temperature, providing high-speed, noise-free convergence for large binary problems. The Digital Annealer supports both exact and heuristic solvers and is capable of handling problem sizes in the thousands of variables range, making it well-suited for grid-scale energy scheduling. Using DADK enables us to evaluate the QUBO formulation rapidly and reliably, offering a practical environment to validate the proposed optimization framework before eventual migration to physical quantum systems.

4.1.1 Problem settings

Next, we define key parameters that frame the optimization problem:

Table 4.1: Battery types and technical characteristics

Type	**Capacity (kWh)**	**Charge (kW)**	**Discharge (kW)**	**C_eff (%)**	**D_eff (%)**	E_0^i **(kWh)**	$E_{\mathbf{req}}^i$ **(kWh)**
EV	200	20	10	90	90	100	60
C&I	2000	200	100	90	90	1000	600
Residential	20	10	5	90	90	10	6

Table 4.2: System-level parameters

Parameter	**Value**
E_0^i (Initial energy per battery)	Defined per battery, e.g., 50% of capacity
E_{req}^i (Required final energy per battery)	Defined per battery, e.g., 30% of capacity
Transformer capacity	10,000 kW
Trading window (T)	8/24 hours
Δt (trading interval)	1 hour

4.1.2 Minimization problem: QUBO formulation

In this section, we introduce the formulation of the problem by presenting the objectives and constraints. Based on this formulation, we propose three optimization strategies, which are summarized in the following tables.

Objectives

Let $b_{i,t}^{(+)} \in \{0,1\}$ and $b_{i,t}^{(-)} \in \{0,1\}$ denote the binary decision variables indicating whether battery i is charging or discharging at time t, respectively. Let η_c^i and η_d^i denote the charge and discharge efficiencies of battery i, and R_c^i, R_d^i be the associated charge/discharge rates (in kW)

Objective 1: Cost Minimization with Market Prices (PnL Maximization/Minimization)

To minimize operational cost, we account for both the energy charged at the market buy price and the energy discharged at the market sell price, across all batteries and time intervals. Let p_t^{buy} and p_t^{sell} denote the electricity buying and selling prices at time t. For each battery i, let η_c^i and η_d^i be its charging and discharging efficiencies, and r_c^i, r_d^i be the charging/discharging power rates.

The cost objective is defined as:

$$\mathcal{H}_{\text{cost}} = \sum_{i=1}^{N} \sum_{t=1}^{T} \left(\eta_c^i \cdot R_c^i \cdot p_t^{\text{buy}} \cdot b_{i,t}^{(+)} - \frac{R_d^i}{\eta_d^i} \cdot p_t^{\text{sell}} \cdot b_{i,t}^{(-)} \right)$$

This objective penalizes charging actions (which incur cost) and rewards discharging actions (which generate revenue) according to their time-dependent market prices.

To ensure comparability with other QUBO components, we normalize this term using the maximum absolute value of its coefficients:

$$\mathcal{H}_{\text{cost}}^{\text{norm}} = \frac{\mathcal{H}_{\text{cost}}}{\max_{i,t} \left(|\eta_c^i \cdot R_c^i \cdot p_t^{\text{buy}}|, \left| \frac{R_d^i}{\eta_d^i} \cdot p_t^{\text{sell}} \right| \right)}$$

Objective 2: Peak-Valley control optimization via Load variance minimization

The goal is to minimize the variance of the net load (flatten peaks and fill valleys). for this we introduce an objective that penalizes the squared value of the net load [1] at each timestep. The net load consists of the exogenous forecasted base load L_t and the contribution of all batteries via their charge and discharge actions.

The method consists of:

[1] Net Load = Base Load + (DERs Charge - DERs Dischage)

- Compute the average net load (L_t^{avg}) over time.
- Penalize deviations from this average $\mathcal{H}_{\text{load}}$.

Let $b_{i,t}^{(+)}$ and $b_{i,t}^{(-)}$ be the binary charge/discharge decisions, and let η_c^i, η_d^i, R_c^i, R_d^i be the efficiencies and power rates of battery i. The average net load (L_t^{avg}) at time t is:

$$L_t^{avg} = \frac{1}{T}\sum_{i=1}^{T}(L_t + \sum_{i=1}^{N}\left(\eta_c^i \cdot R_c^i \cdot b_{i,t}^{(+)} - \frac{R_d^i}{\eta_d^i} \cdot b_{i,t}^{(-)}\right))$$

To control the peak-shaving and valley-filling by minimizing the load variance, we define the following squared penalty:

$$\mathcal{H}_{\text{load}} = \sum_{t=1}^{T}\left(\frac{1}{\gamma} \cdot \left(\sum_{i=1}^{N}\left(\eta_c^i \cdot R_c^i \cdot b_{i,t}^{(+)} - \frac{R_d^i}{\eta_d^i} \cdot b_{i,t}^{(-)}\right) + L_t - L_t^{avg}\right)\right)^2$$

where $\gamma = \max_t L_t + 1$ is a normalization factor used to scale the penalty term and ensure numerical stability within the QUBO formulation.

This objective encourages DERs to schedule operations in a way that flattens load profiles and avoids peak transformer usage.

Objective 3: Battery Readiness

To encourage each battery i to end the trading horizon with an energy level close to a target requirement E_{req}^i, we formulate a soft penalty based on squared deviation from this target. Let E_0^i denote the initial energy in battery i. The cumulative net energy added (charging minus discharging) over time is computed by summing energy inflows/outflows at each timestep t and its preceding intervals $v \leq t$.

The battery readiness objective is defined as:

$$\mathcal{H}_{\text{energy}} = \sum_{i=1}^{N}\left(\sum_{t=1}^{T}\sum_{v=1}^{t}\left(\eta_c^i \cdot R_c^i \cdot b_{i,v}^{(+)} - \frac{R_d^i}{\eta_d^i} \cdot b_{i,v}^{(-)}\right) + E_0^i - E_{\text{req}}^i\right)^2$$

This term penalizes batteries that fail to meet the minimum final energy threshold, while allowing flexibility in how and when energy is charged/discharged during the trading window.

We normalize this term as follows:

$$\mathcal{H}_{\text{energy}}^{\text{norm}} = \frac{1}{T \cdot N \cdot (\max_i |E_0^i - E_{\text{req}}^i| + 1)^2} \cdot \mathcal{H}_{\text{energy}}$$

Constraints

Constraint 1: Charging/Discharging Exclusivity (Hard)

To ensure that no battery charges and discharges simultaneously at any time step, we enforce a hard exclusivity constraint between the binary decision variables $b_{i,t}^{(+)}$ and $b_{i,t}^{(-)}$. Violations are penalized in the QUBO formulation through a quadratic term:

$$\mathcal{H}_{\text{exclusive}} = \sum_{i=1}^{N}\sum_{t=1}^{T} b_{i,t}^{(+)} \cdot b_{i,t}^{(-)}$$

This term penalizes configurations where both $b_{i,t}^{(+)}$ and $b_{i,t}^{(-)}$ are set to 1 at the same timestep t for battery i. As this is a hard constraint, the associated penalty weight in the final QUBO formulation should be set sufficiently high to prevent violation.

To balance it numerically with other terms in the QUBO model, we apply a normalization factor:

$$\mathcal{H}_{\text{exclusive}}^{\text{norm}} = \frac{1}{N \cdot T} \cdot \mathcal{H}_{\text{exclusive}}$$

Constraint 2: No Consecutive Charging or Discharging Periods (Soft)

To discourage continuous cycling and preserve battery flexibility and health, we penalize patterns where a battery charges or discharges for three or more consecutive time steps. For every battery i and time window $[t, t+1, t+2]$, we define a soft penalty based on the sum of the respective binary decision variables.

The penalty for consecutive charging is defined as:

$$\mathcal{H}_{\text{consec}}^{(c)} = \sum_{i=1}^{N} \sum_{t=1}^{T-3} \left(b_{i,t}^{(+)} + b_{i,t+1}^{(+)} + b_{i,t+2}^{(+)} - 2 \right)^2$$

Similarly, the penalty for consecutive discharging is:

$$\mathcal{H}_{\text{consec}}^{(d)} = \sum_{i=1}^{N} \sum_{t=1}^{T-3} \left(b_{i,t}^{(-)} + b_{i,t+1}^{(-)} + b_{i,t+2}^{(-)} - 2 \right)^2$$

These expressions penalize sequences where three consecutive charging or discharging actions occur, while allowing up to two such actions in a 3-step window. This constraint is soft, meaning it is included in the total QUBO cost function with a tunable penalty weight.

We normalize the penalties by the number of terms as follows:

$$\mathcal{H}_{\text{consec}}^{\text{norm}} = \frac{1}{N \cdot (T-2)} \cdot \left(\mathcal{H}_{\text{consec}}^{(c)} + \mathcal{H}_{\text{consec}}^{(d)} \right)$$

Constraint 3: Transformer Capacity Limit (Hard)

To prevent overloading the transformer, we impose a constraint ensuring that the aggregated battery operations at each timestep t do not exceed the available capacity K_t, defined as the difference between the transformer's rated limit and the exogenous load at time t.

Let η_c^i, η_d^i be the charge/discharge efficiencies of battery i, and R_c^i, R_d^i be the corresponding rates. The total contribution of all batteries to the net load at time t is given by:

$$L_t^{\text{bat}} = \sum_{i=1}^{N} \left(\eta_c^i \cdot R_c^i \cdot b_{i,t}^{(+)} - \frac{R_d^i}{\eta_d^i} \cdot b_{i,t}^{(-)} \right)$$

We penalize any configuration where $L_t^{\text{bat}} > K_t$ through the following hard constraint:

$$\mathcal{H}_{\text{transformer}} = \sum_{t=1}^{T} \left(L_t^{\text{bat}} - K_t \right)^2$$

This constraint strictly enforces that net battery load stays below the transformer's headroom at every timestep.

To balance this term within the QUBO framework, we normalize it using the square of the maximum observed capacity margin:

$$\mathcal{H}_{\text{transformer}}^{\text{norm}} = \frac{1}{\max_t K_t^2} \cdot \mathcal{H}_{\text{transformer}}$$

Constraint 4: Final SOC Requirement (Soft)

To ensure that each battery retains sufficient energy at the end of the trading window, we introduce a soft constraint requiring that the total net energy retained after all charging and discharging operations is at least $\beta\%$ of its nominal capacity.

Let Capacity_i be the energy capacity of battery i. The net energy accumulated during the trading horizon $[1, T]$ is:

$$E_{\text{net}}^i = \sum_{t=1}^{T} \left(\eta_c^i \cdot R_c^i \cdot b_{i,t}^{(+)} - \frac{R_d^i}{\eta_d^i} \cdot b_{i,t}^{(-)} \right)$$

We define a slack-based penalty to enforce:

$$E_{\text{net}}^i \geq 0.3 \cdot \text{Capacity}_i$$

This inequality is transformed into a QUBO-compatible soft constraint using a squared penalty:

$$\mathcal{H}_{\text{SOC}} = \sum_{i=1}^{N} \frac{1}{\text{Capacity}_i^2} \left(E_{\text{net}}^i - 0.3 \cdot \text{Capacity}_i + s_i\right)^2$$

where s_i is a slack variable introduced to maintain QUBO compatibility. This constraint encourages—but does not strictly enforce—final energy availability per battery, preserving flexibility for economic optimization.

Full QUBO Formulation

The optimization problem is defined as a Quadratic Unconstrained Binary Optimization (QUBO) model over binary decision variables $b_{i,t}^{(+)}$ and $b_{i,t}^{(-)}$, representing the charging and discharging actions of battery i at time t.

We aim to minimize the following composite objective function:

$$\begin{aligned} \min_{\{b_{i,t}^{(+)}, b_{i,t}^{(-)}\}} \quad & \mathcal{H}_{\text{total}} = \lambda_1 \cdot \mathcal{H}_{\text{cost}}^{\text{norm}} + \lambda_2 \cdot \mathcal{H}_{\text{load}} + \lambda_3 \cdot \mathcal{H}_{\text{readiness}}^{\text{norm}} \\ & \quad + \lambda_4 \cdot \mathcal{H}_{\text{exclusive}}^{\text{norm}} + \lambda_5 \cdot \mathcal{H}_{\text{consec}}^{\text{norm}} + \lambda_6 \cdot \mathcal{H}_{\text{transformer}}^{\text{norm}} + \lambda_7 \cdot \mathcal{H}_{\text{SOC}}^{\text{norm}} \end{aligned}$$

where:

- $\mathcal{H}_{\text{cost}}^{\text{norm}}$ is the normalized cost minimization objective,
- $\mathcal{H}_{\text{load}}$ is the load peak-shaving valley-filling objective,
- $\mathcal{H}_{\text{readiness}}^{\text{norm}}$ is the normalized battery readiness penalty,
- $\mathcal{H}_{\text{exclusive}}^{\text{norm}}$ is the exclusivity constraint ensuring no simultaneous charge/discharge,
- $\mathcal{H}_{\text{consec}}^{\text{norm}}$ penalizes 3-step consecutive charging/discharging,
- $\mathcal{H}_{\text{transformer}}^{\text{norm}}$ enforces the transformer capacity limit,
- $\mathcal{H}_{\text{SOC}}^{\text{norm}}$ penalizes violations of the minimum final SOC requirement.

Each term is associated with a weight λ_j to tune the relative importance of the objective and constraint components in the QUBO solution.

Table 4.3: Optimization strategies and their objectives, constraints, and penalization weights

Strategy	Objective(s)	Constraint(s)	Penalization Factors
1	• Maximize profit versus loss	• Exclusivity • Consecutive charging • Consecutive discharging • Final SOC required • Transformer capacity limit	• λ_{cost}: 100 • $\lambda_{\text{exclusivity}}$: 20 • $\lambda_{\text{consec_charg}}$: 1 • $\lambda_{\text{consec_discharg}}$: 1 • λ_{soc}: 5 • $\lambda_{\text{transformer}}$: 5
2	• Maximize profit versus loss • Maximize battery readiness	• Exclusivity • Consecutive charging • Consecutive discharging • Transformer capacity limit	• λ_{cost}: 50 • $\lambda_{\text{readiness}}$: 300 • $\lambda_{\text{exclusivity}}$: 10 • $\lambda_{\text{consec_charg}}$: 1 • $\lambda_{\text{consec_discharg}}$: 1 • $\lambda_{\text{transformer}}$: 5
3	• Maximize profit versus loss • Maximize battery readiness • Peak shaving	• Exclusivity • Consecutive charging • Consecutive discharging • Transformer capacity limit	• λ_{cost}: 10 • $\lambda_{\text{readiness}}$: 200 • λ_{load}: 100 • $\lambda_{\text{exclusivity}}$: 10 • $\lambda_{\text{consec_charg}}$: 1 • $\lambda_{\text{consec_discharg}}$: 1 • $\lambda_{\text{transformer}}$: 5

4.1.3 Use cases

In this section, we present the results of applying the previously formulated optimization problem to two data sets: Synthetic data set (with 5 DERs and 8 Time intervals) and Real case (with 15 DERs and 24 Time Intervals) where we use the California Independent System Operator (CAISO) NP15 Zonal Market Data as input data. The synthetic data are applied to 3 strategies separately (Tables 4.3 and 4.6), where strategy 3 is applied to the real data [2].

Performance Metrics

We define the following performance metrics to evaluate the outcome of each optimization strategy:

To compare results in energy terms, all power-based operations are scaled by the time interval Δt, expressed in hours.

1. Profit and Loss (PnL): The net profit is the total revenue from energy sold minus the cost of energy purchased, accounting for time scaling:

$$\text{PnL} = \sum_{i=1}^{N} \sum_{t=1}^{T} \Delta t \cdot \left(\frac{R_d^i}{\eta_d^i} \cdot p_t^{\text{sell}} \cdot b_{i,t}^{(-)} - \eta_c^i \cdot R_c^i \cdot p_t^{\text{buy}} \cdot b_{i,t}^{(+)} \right)$$

2. Battery Energy Exchange (BEE): Net energy exchange is computed as the difference between charged and discharged energy:

$$\text{BEE} = \sum_{i=1}^{N} \sum_{t=1}^{T} \Delta t \cdot \left(\frac{R_d^i}{\eta_d^i} \cdot b_{i,t}^{(-)} - \eta_c^i \cdot R_c^i \cdot b_{i,t}^{(+)} \right)$$

3. Battery System Energy Evolution (BSEE): The state-of-charge (SOC) of battery i at time t is expressed as the initial energy plus the net accumulated energy scaled by Δt, normalized by the battery capacity:

$$\text{SOC}_{i,t} = \frac{E_0^i + \sum_{v=1}^{t} \Delta t \cdot \left(\eta_c^i \cdot R_c^i \cdot b_{i,v}^{(+)} - \frac{R_d^i}{\eta_d^i} \cdot b_{i,v}^{(-)} \right)}{\text{Capacity}_i}$$

The trajectory of $\text{SOC}_{i,t}$ over all time steps $t = 1, \ldots, T$ defines the Battery System Energy Evolution (BSEE). where:

- E_0^i is the initial energy (kWh) of battery i,
- Δt is the trading interval in hours,
- R_c^i, R_d^i are the charging/discharging rates (kW),
- η_c^i, η_d^i are the charge/discharge efficiencies,
- $b_{i,v}^{(+)}, b_{i,v}^{(-)} \in \{0, 1\}$ are the binary decisions at time v.

This expression tracks the normalized SOC trajectory of each battery over time, incorporating both physical constraints and operational decisions.

Synthetic dataset

[2]Note: Coloring code for Charging/Discharging scheduling figures is Red (1) = Charge, Blue (-1) = Discharge, Black (0) = Idle.

Table 4.4: Algorithmic Response to Electricity Demand Levels

Demand Level	**Algorithm Action**	**Operation**	**Result**
High base load	Discharge DERs	Base load - Discharge (return power to grid)	Peak shaving
Low base load	Charge DERs	Base load + Charge (draw power from grid)	Valley filling

Table 4.5: Algorithmic Response to Electricity Price Signals

Input (Electricity Price)	**Algorithm Action**	**Output**
Sell price high	Discharge	Revenue increases
Sell price low	Idle	Revenue constant
Buy price high	Idle	Cost constant
Buy price low	Charge	Cost increases

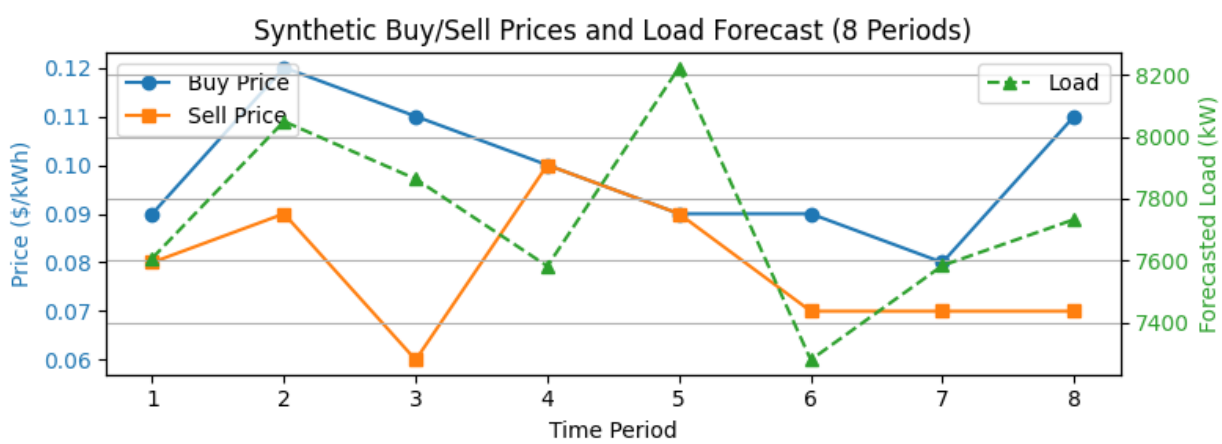
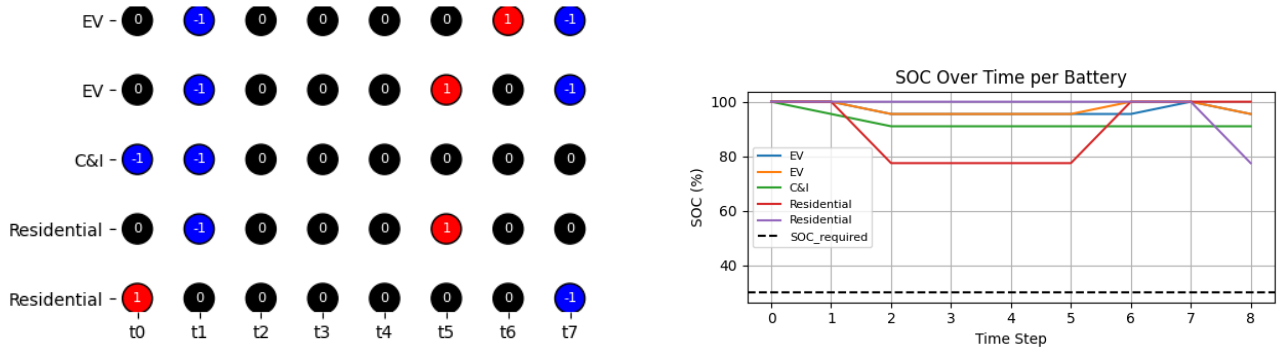


Figure 4.1: Synthetic dataset.

Table 4.6: Summary of Strategy Results

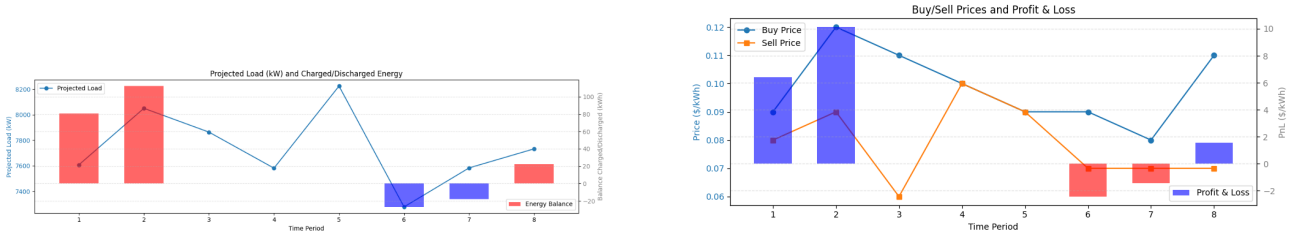
Strategy	**C/D Schedule**	**PnL Evolution**	**BEE Evolution**	**BSEE (SOC Evolution)**	**PnL**	**BEE**
1	Fig. 4.2a	Fig. 4.3b	Fig. 4.3a	Fig. 4.2b	14.22$	171 kWh
2	Fig. 4.4a	Fig. 4.5b	Fig. 4.5a	Fig. 4.4b	-10$	-126 kWh

Strategy 1



(a) Charging/Discharging scheduling (b) SOC evolution per battery

Figure 4.2: Plots for strategy 1: C/D scheduling and SOC evolution



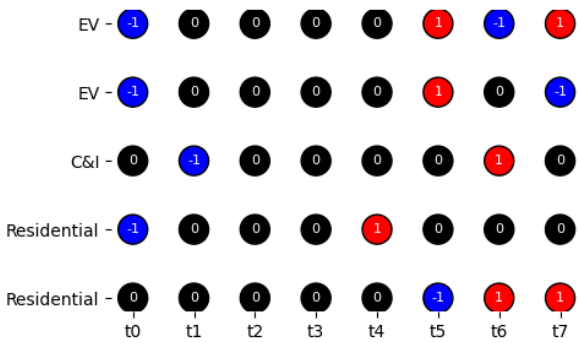
(a) Electricity load vs Balance Exchange Energy (b) Electricity price vs PnL

Figure 4.3: Plots for strategy 1: BEE and PnL

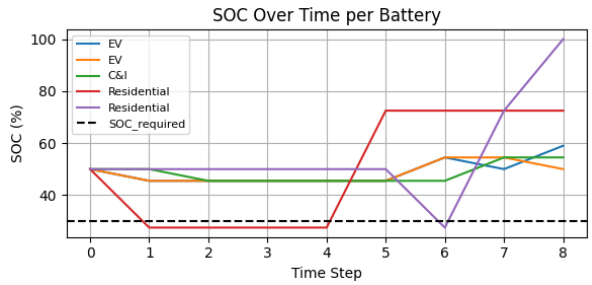
Interpretation

When electricity sell prices are high, the algorithm is expected to favor discharging behavior. However, this response is only partially observed due to the influence of the final state-of-charge constraint ($\text{SOC}_{\text{final}} > 30\%$), which dominates the charging and discharging schedule. As a result, all batteries tend to finish the trading window with a state of charge close to 100%.

Strategy 2

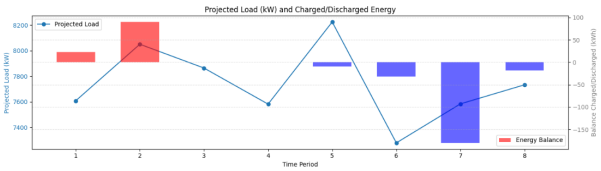


(a) Charging/Discharging scheduling



(b) SOC evolution per battery

Figure 4.4: Plots for strategy 2: C/D scheduling and SOC evolution



(a) Electricity load vs Balance Exchange Energy

(b) Electricity price vs PnL

Figure 4.5: Plots for strategy 2: BEE and PnL

Interpretation

A similar behavior to that observed in Strategy 1 emerges here, where discharging actions are triggered when electricity buy prices are low. In this strategy, the optimization favors maintaining the state of charge close to the required threshold throughout the trading window, rather than only at its conclusion. The Load vs. BEE plot further illustrates that during periods of low electricity load, the algorithm effectively discharges energy to the grid, thereby shaving the load curve as intended.

Strategy 3

Interpretation

Table 4.7: Comparison of DER Scheduling vs. No Scheduling

Feature	**DERs Scheduling**	**DER No-Scheduling**
Reduce trading cost	✓	✗
Increases trading revenue	✓	✗
Peak shaving (Reduces electrical congestion)	✓	✗
Valley filling (Absorbs excess generated energy)	✓	✗
Battery readiness	✓	✓
Computing time (AQC)	✓	✗
Computing cost (large scale)	✗	✗
Public server usage	✗	✗

Strategy 3 incorporates peak shaving and valley filling as additional objectives. The algorithm effectively addresses all three goals—maximizing revenue (with a cumulative PnL of 6$) by discharging during periods of high sell prices, moderately shaving demand during peak periods, and flattening the projected electricity load during low-demand intervals (achieving a Peak-Valley reduction of 36%). This charging and discharging behavior affects the evolution of the batteries' state of charge (SOC), maintaining SOC levels close to the required energy threshold (30%) for all batteries except one, which shows a slight deviation from the intended target.

Table 4.8: Summary of Strategy 3 on Synthetic dataset

Strategy	**Profit and Loss analysis**	**Base load vs Base Load + DERs**	**Overall battery readiness**	**Load vs PnL**
3	Fig. 4.7a	Fig. 4.7c	Fig. 4.7g	Fig. 4.7b

Table 4.9: Summary 2 of Strategy 3 on Synthetic dataset

Strategy	**C/D Schedule**	**Buy Price vs Charging Cost**	**Sell Price vs Discharging Revenue**	**C/D Schedule vs Load**	**BSEE (SOC Evolution)**	**PnL**
3	Fig. 4.6	Fig. 4.7e	Fig. 4.7f	Fig. 4.7d	Fig. 4.7h	6$

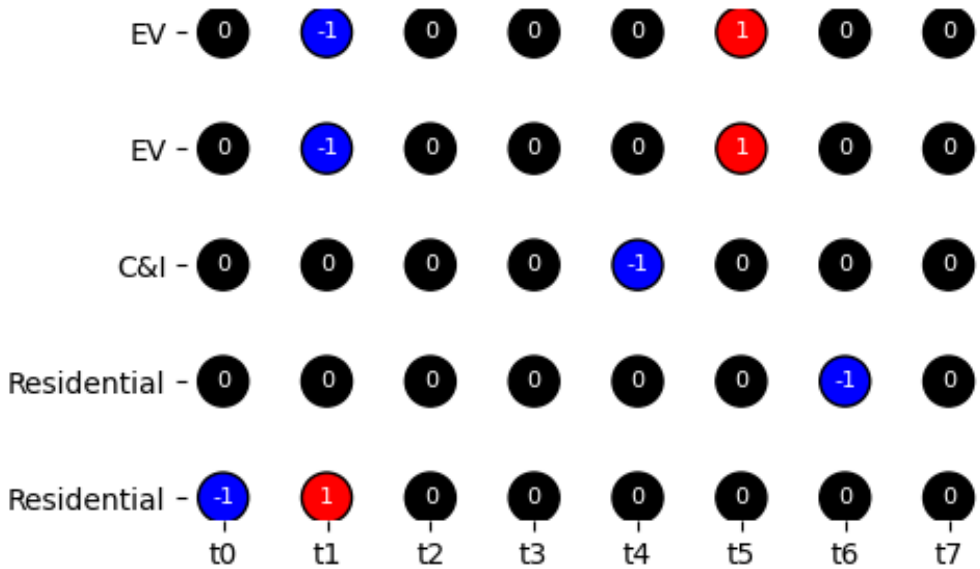


Figure 4.6: Charging and Discharging schedule synthetic data.

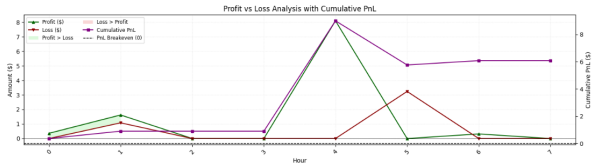
Real dataset: CAISO NP15 Zonal Market Data

Table 4.10: Summary of Strategy 3 on CAISO NP15 Zonal Market Data

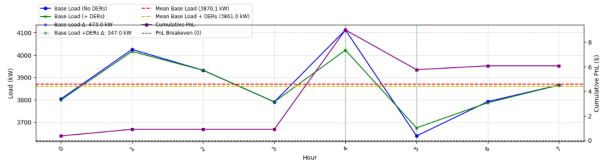
Strategy	**Profit and Loss analysis**	**Base load vs Base Load + DERs**	**Overall battery readiness**	**Load vs PnL**
3	Fig. 4.10a	Fig. 4.10b	Fig. 4.10c	Fig. 4.10d

Table 4.11: Summary 2 of Strategy 3 on CAISO NP15 Zonal Market Data

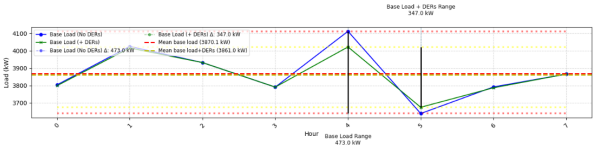
Strategy	**C/D Schedule**	**Buy Price vs Charging Cost**	**Sell Price vs Discharging Revenue**	**C/D Schedule vs Load**	**BSEE (SOC Evolution)**	**PnL**
3	Fig. 4.9	Fig. 4.11b	Fig. 4.11c	Fig. 4.11a	Fig. 4.11d	30.50$



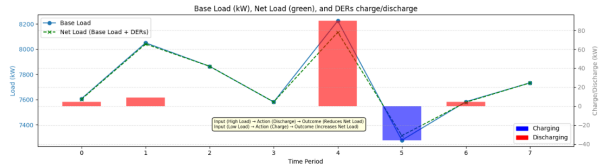
(a) PnL



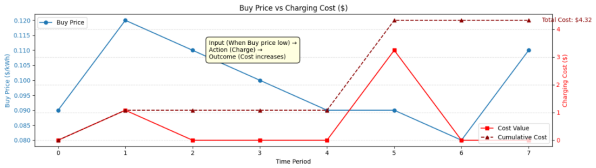
(b) Base Load + DERs vs PnL



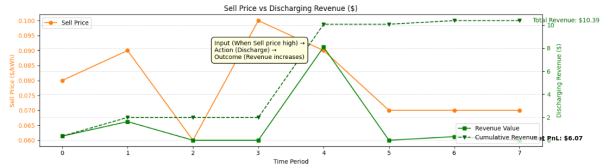
(c) Base Load + DERs



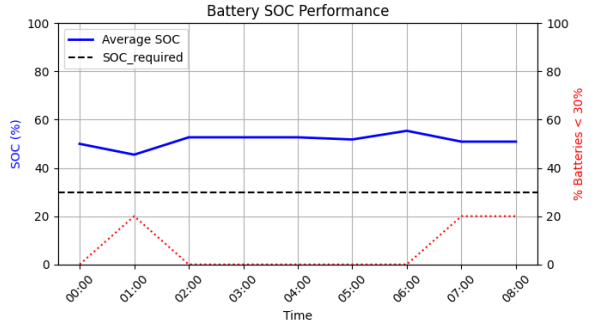
(d) Base Load + DERs vs Charging and Discharging



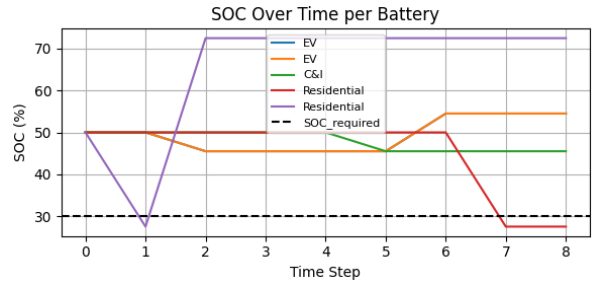
(e) Buy price vs charging cost



(f) Sell price vs discharging revenue



(g) Battery readiness



(h) SOC evolution per battery

Figure 4.7: Combined plots for strategy 3 using synthetic data

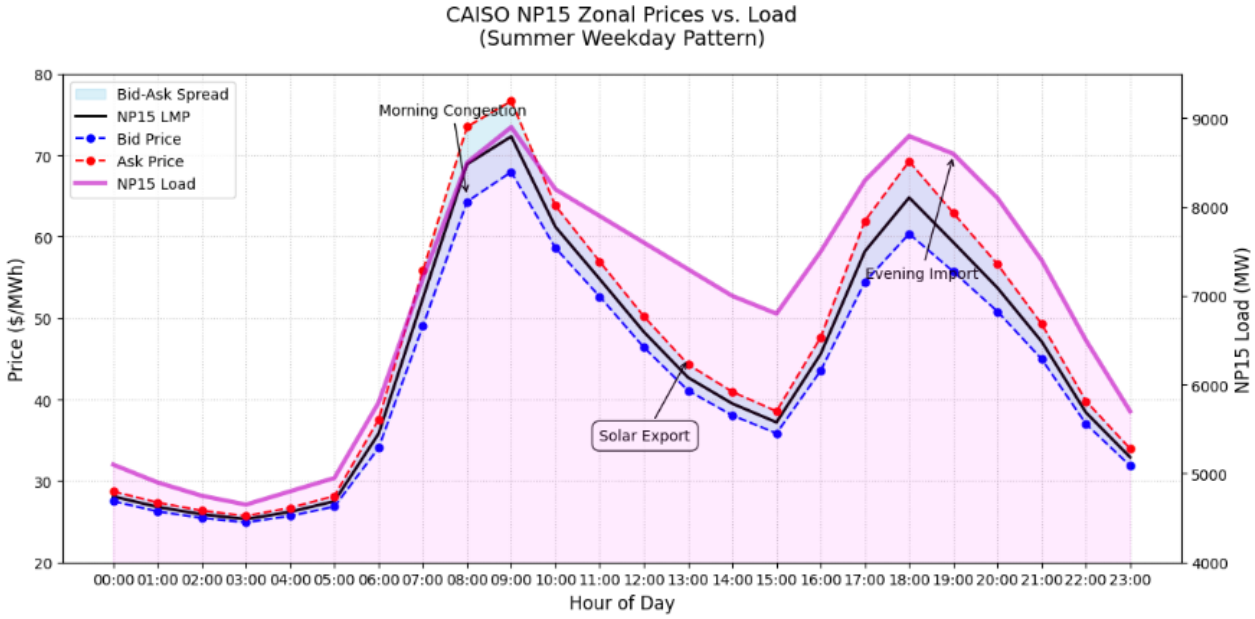


Figure 4.8: CAISO NP15 Zonal Market Data.

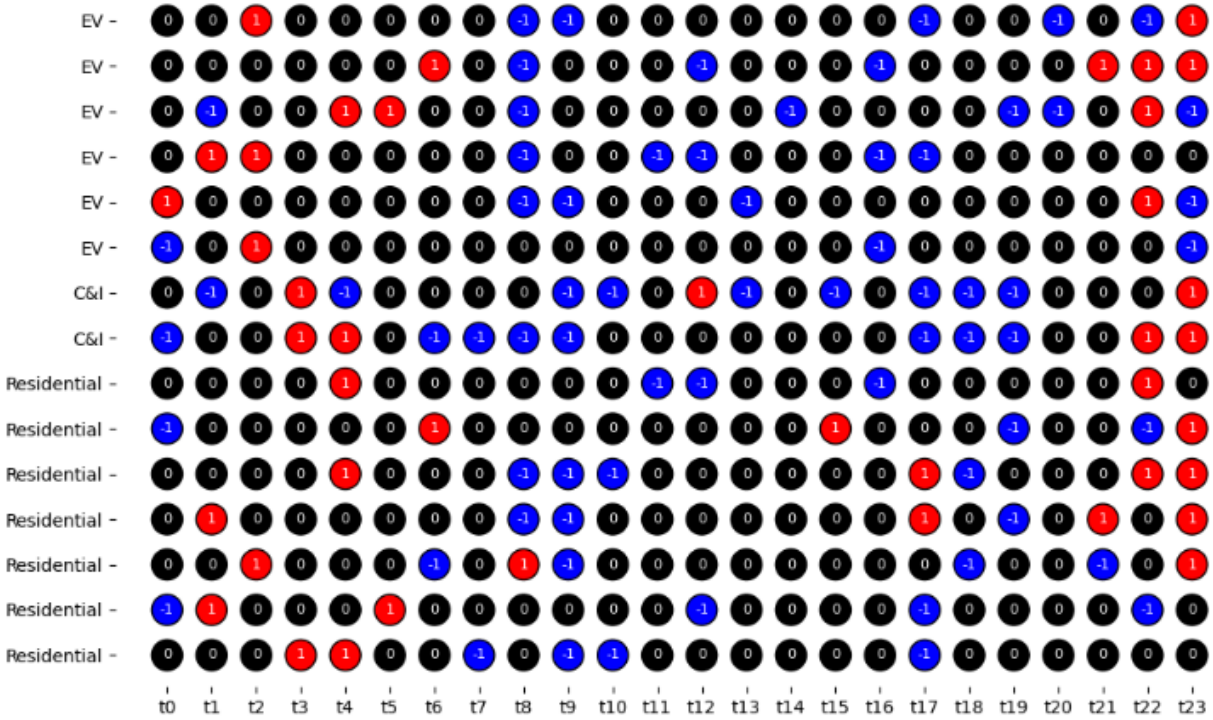
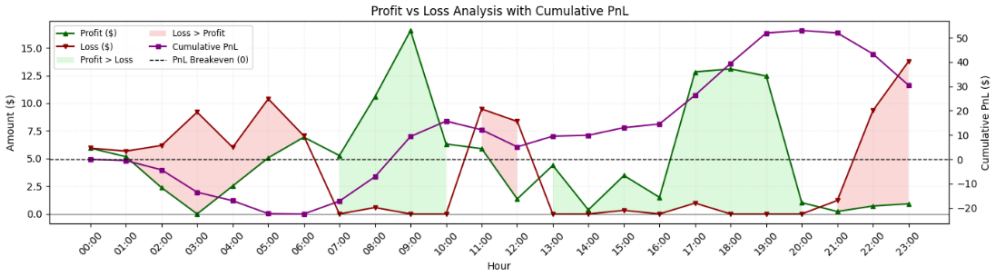
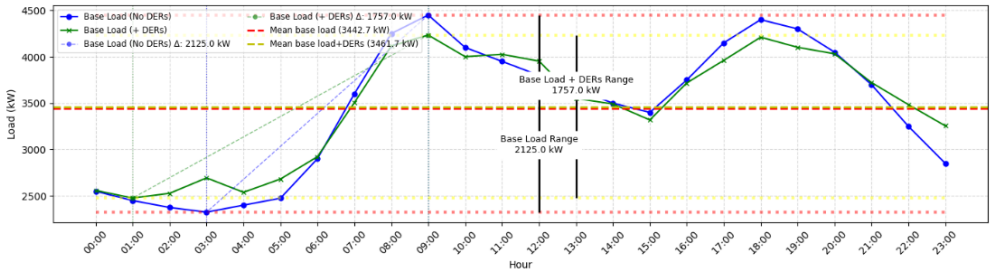


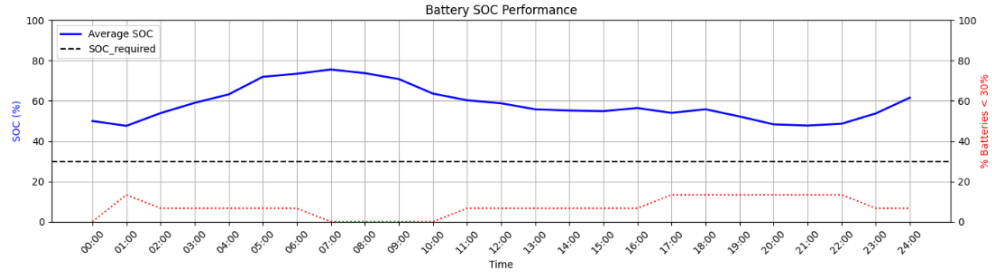
Figure 4.9: Charging/Discharging scheduling



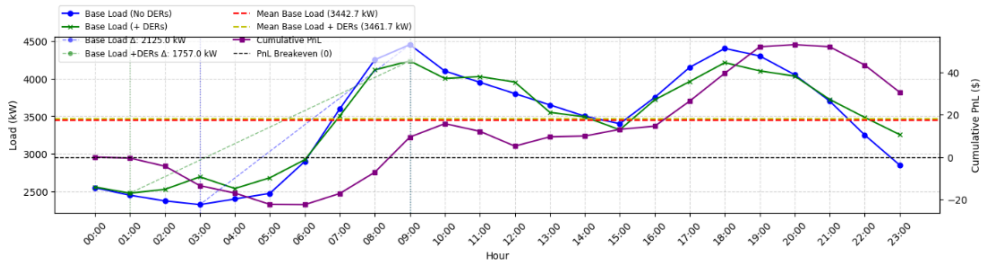
(a) Profit and Loss analysis



(b) Base load vs Base load + DERs

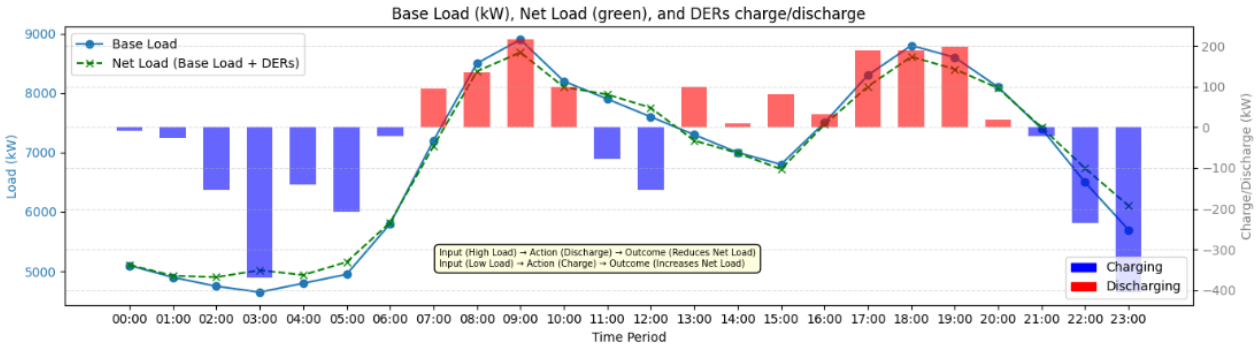


(c) Overall battery readiness

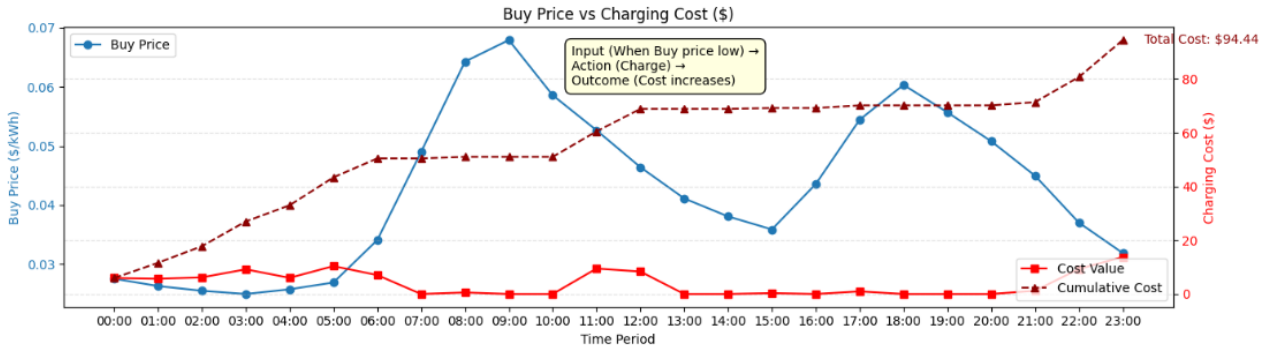


(d) Net load vs PnL

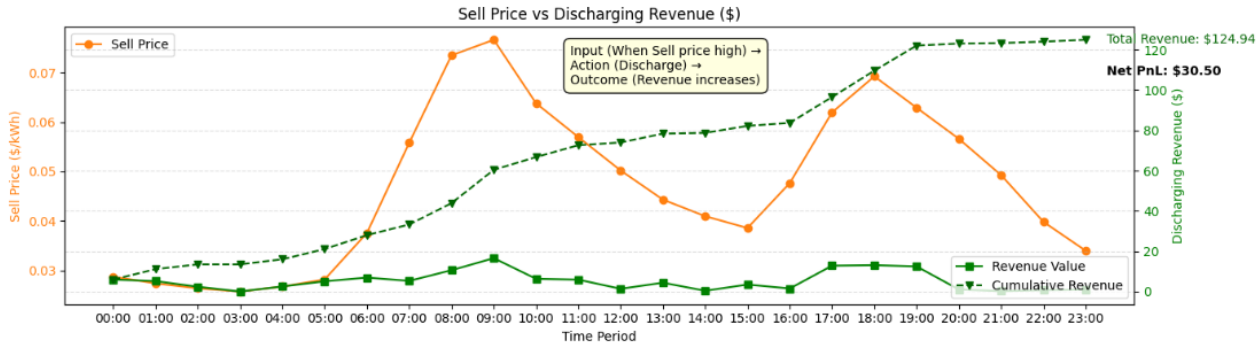
Figure 4.10: Combined plots for Strategy 3: Overall results



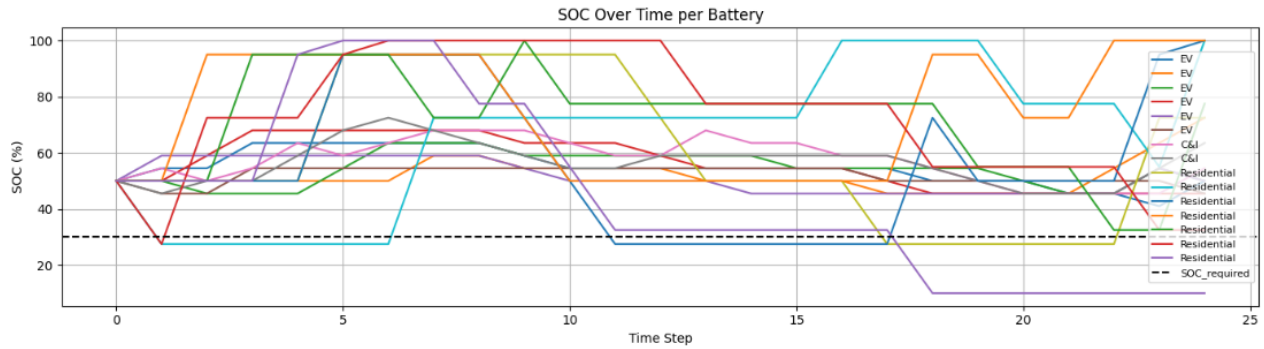
(a) Projected base load vs Net load = Base load + DERs



(b) Electricity buy price vs charging cost



(c) Electricity sell price vs discharging revenue



(d) SOC evolution per battery

Figure 4.11: Combined plots for Strategy 3

Interpretation

Figure 4.10a presents the cumulative Profit and Loss (PnL) evolution, highlighting the specific time periods when the algorithm generates profits versus losses. Over the 24-hour trading window, the algorithm achieves a net profit of $30. While this amount may appear modest, it represents significant savings when compared to the baseline scenario without battery optimization and all incurred costs in unmanaged operations.

Figure 4.10b and table 4.12 illustrate the impact of battery charging and discharging behavior on the total electricity load over time with the goal of Peak-Valley control optimization. As expected from the algorithm's design (see Table 4.4), when electricity load is high, the system tends to discharge in order to shave the peak, hence level up the base load (base load - DERs discharge). Contrary, when electricity load is low, the algorithm tends to charge in order to fill up the valley. This helps reduce the difference between peak and valley before and after the DERs operations: the delta of Base load (No DERs) is 2150 kW, while the delta of Base load + DERs is reduced to 1757 kW. This corresponds to a peak-valley reduction of 23%.

Table 4.12: Comparison of Peak-Valley (PV) Load With and Without DER Control

Case	**Peak-Valley Time**	**Net Load**	**Delta (kW)**
Without PV Control (No DERs)	Peak (6PM), Valley (3AM)	Peak: 4400 Valley: 2250	2150
With PV Control (+ DERs)	Peak (6PM), Valley (3AM)	Peak: 4257 Valley: 2500	1757

Figure 4.10c presents the overall state-of-charge (SOC) evolution across the battery ecosystem, along with the percentage of batteries that do not meet the required SOC threshold for emergency readiness at each time step. The algorithm maintains relatively stable SOC behavior, consistently hovering around 60%.

Finally, Figure 4.10d overlays the electricity demand and the PnL evolution. It is observed that PnL remains relatively flat when demand is low, but increases significantly during periods of high demand. This behavior reflects the algorithm's response to discharge when demand is high, effectively leveling the total demand and generating positive revenue for the battery system owners.

Figure 4.11a illustrates the operational dynamics of Distributed Energy Resources (DERs) in relation to the projected electricity load profile. The figure demonstrates how coordinated charging and discharging of DERs achieves effective peak shaving (reducing maximum demand) and valley filling (absorbing excess generation), resulting in a flattened net load curve.

Figure 4.11b captures the charging costs (losses) in relation to the electricity buy price. High buy prices are interpreted by the algorithm as signals to remain idle—neither charging nor discharging—while low buy prices are considered favorable opportunities for charging, which results in a cost, albeit an optimized one (see Table 4.5).

Figure 4.11c illustrates the algorithm's behavior in response to electricity sell prices. Periods of high sell prices are met with battery discharging, thus maximizing revenue.

Finally, Figure 4.11d presents the individual state-of-charge (SOC) evolution for each battery in the system.

Chapter 5

Computating Time

5.1 Computating Time

All computations were run using the following system: Intel(R) Core(TM) i7-8550U CPU @ 1.80GHz, 1992 MHz, 16GB RAM. The number of variables for each run is giving by (Table 5.1 and Figure 5.1):

$$2 \times (N_{\text{DERs}} \times T + 3 \times N_{\text{DERs}})$$

Where:
N_{DERs} — Number of Distributed Energy Resources (DERs)
T — Number of time periods

We see a linear relationship between the number of DERs (number of variables) included in the model and the computating time of the quantum algorithm.

Table 5.1: Computation Time vs. Number of Variables

Number of Variables (DERs)	Computing Time (sec)
110 (5)	32
220 (10)	69
330 (15)	111
550 (25)	200
1100 (50)	422
2200 (100)	1040
3300 (150)	1595

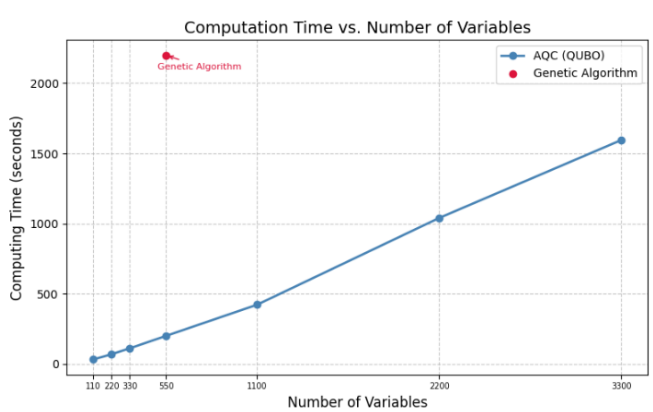


Figure 5.1: Computation scalability

Chapter 6

Conclusions and future work

6.1 Conclusions and future work

This work demonstrates the potential of Adiabatic Quantum Computing (AQC) as a promising approach for solving combinatorial optimization problems in the energy domain. By applying a QUBO-based formulation to a battery trading strategy, we show that the algorithm performs in line with expectations, effectively balancing objectives such as profit maximization, load leveling, and energy availability for emergency readiness.

This Proof of Concept (PoC) successfully shows a practical Energy Trading implementation, yielding two key insights:

1. The quantum approach generates highly optimal trading solutions
2. The system scales linearly with problem size, unlike classical methods that exhibit exponential complexity growth

These results validate Quantum Computing as a transformative technology for large-scale energy optimization problems.

Future work will focus on several key areas: comparing more classical and quantum optimization algorithms; implementing the forecasting and classification modules within the QET framework using classical and/or quantum machine learning; expanding the model to include additional Distributed Energy Resources (DERs) such as Solar Panels; enhancing the realism of the formulation by modeling initial SOC based on actual charging behavior; and incorporating user-defined preferences for required energy reserves. Additionally, to support transparency and reproducibility, the complete code implementation is publicly available in the dedicated GitHub repository (https://github.com/ismaelahrazem/Quantum-Computing-for-Industry-Practitioners).

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